

Vitamin D: 1. supplements and testing

National Osteoporosis Society 2018, Vitamin D and Bone Health, Vitamin D and health, SACN, July 2016, CMO guidance 2012, NICE 2025 NG247, BMJ 2022;378:e070270, NICE 2017, PH56, BMJ 2010;340:b5664



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Who needs ROUTINE vitamin D supplementation (even if not deficient)?

It is easiest to remember this as 2 groups:

- People with insufficient sun exposure year-round to make adequate vitamin D – and everyone else...

Insufficient sun exposure

- Individuals with low sun exposure, e.g. institutionalised, skin always covered
- Darker skin (need higher levels of sun exposure to synthesise enough vitamin D)

Vitamin D supplements all year:
400 IU (10mcg)/d;
if <1y, see below

Everyone else

Over winter in the UK, we should ALL consider supplementing vitamin D due to low sun exposure, but some with higher requirements should be supplemented year-round.

<1y <i>If breastfed or if formula fed and <500ml of formula/day</i>	1–4y	5–64y <i>(except pregnancy/breastfeeding; see below)</i>	≥65y
Vitamin D all year 8.5–10mcg daily	Vitamin D all year 400 IU (10mcg)/d	<i>Consider</i> 400 IU (10mcg)/d from October to early March	Vitamin D all year 400 IU (10mcg)/d

Pregnancy and breastfeeding: what about routine supplementation? NICE 2024, NG247 recommends that pregnant and breastfeeding women take 10mcg (400IU) per day between October and March, unless they are in the insufficient sun exposure group, which needs year-round supplements.

Other factors that increase the risk of vitamin D deficiency: testing/supplementing may be indicated

- Medical conditions, e.g.:
 - Coeliac disease: may need calcium and vitamin D supplements if dietary intake insufficient (NICE NG20, 2015).
 - After bariatric surgery: see the article *Bariatric surgery* for details.
- Drugs, e.g.:
 - Antiepileptic drugs: see the article *Epilepsy* for details.
 - Orlistat: may impair absorption; consider vitamin D supplementation (BNF, accessed April 2023).

Who needs vitamin D TESTING? (the test of choice is a 25OH vitamin D level)

Testing suggested if:	Do NOT need testing:
National Osteoporosis Society recommends testing adults if: • Symptoms suggestive of vitamin D deficiency (see table below). • Bone disease that may be improved with vitamin D treatment, e.g. osteomalacia or osteoporosis as part of diagnostic work-up. • Bone disease and starting treatment that requires correction of vitamin D levels (potent antiresorptive treatments, e.g. zoledronate or denosumab, but not oral bisphosphonates); this is because of the risk of hypocalcaemia. A BMJ review suggested the following groups as well: • Hyperparathyroidism, hypophosphataemia, hypo/hypercalcaemia. • Malabsorption syndromes. • Chronic kidney disease (usually only stages 4 and 5). • On medication that increases the risk of vitamin D deficiency, e.g. anticonvulsants, antifungals, antiretrovirals. • Isolated raised ALP (may be due to vitamin D deficiency).	• Asymptomatic healthy (population-risk) individuals. • Asymptomatic individuals at high risk of vitamin D deficiency (because they should be on supplements!). • Those with known osteoporosis/fragility fracture taking regular vitamin D (unless they develop clinical features suggestive of vitamin D deficiency).

Symptoms of vitamin D deficiency (symptoms are often vague, but include the following)

- **Rickets (children):** tetany, irritability, leg bowing or knock knees, impaired growth, skeletal deformity, muscle pain/weakness, bone pain and proximal myopathy. **REFER!**
- **Osteomalacia (adults):** bone pain/tenderness/myalgia, proximal muscle weakness, insufficiency fractures AND no other plausible explanation. **Follow local guidance (usually treatment in primary care and refer if not responding to treatment/concerned for other reasons).**

Vitamin D: 2. Treatment of vitamin D deficiency

National Osteoporosis Society 2018, Vitamin D and Bone Health, Vitamin D and health, SACN, July 2016, <https://www.vegansociety.com>, BMJ 2010;340:b5664



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This applies to ADULTS only (and excludes pregnant women and people with CKD 4 and 5).

This is summarised from the National Osteoporosis Society guidance on Vitamin D and Bone Health, 2018.

Local pathways may vary.

If someone is found to be vitamin D deficient, think about levels in the rest of the family.

Interpreting vitamin D levels

<25nmol/L

25–50nmol/L

>50nmol/L

Vitamin D deficient
Treat

May be inadequate: treat if ≥ 1 of:

- Symptoms of vitamin D deficiency
- Osteoporosis
- Fragility fracture
- High fracture risk
- Drug treatment for bone disease
- Increased risk of vitamin D deficiency, e.g.
 - Reduced sun exposure
 - Raised PTH
 - Drugs: anticonvulsants, glucocorticoids
 - Malabsorption: e.g. coeliac, IBD

Vitamin D sufficient

- No treatment indicated
- Maintain vitamin D by safe sun exposure and diet
- Maintenance vitamin D OTC if appropriate, as per Public Health guidance

Treatment of vitamin D deficiency

Oral vitamin D3 (cholecalciferol) is the replacement and maintenance treatment of choice. Cholecalciferol supplements can be derived from an animal source, and some people may wish to avoid this for religious, ethical or dietary reasons.

The Vegan Society provides information on vitamin D3 derived from lichen.

Annual depot injection of vitamin D is NOT recommended: concerns about unpredictable bioavailability and toxicity.

SYMPTOMATIC
(vitamin D deficiency with symptoms)

ABOUT TO START POTENT ANTIRESORPTIVE AGENT
(zoledronate, denosumab or teriparatide – but not oral treatment)

EVERYONE ELSE
with vitamin D deficiency

- **Correct vitamin D rapidly:** approx. 300 000 IU vitamin D orally in divided doses over 6–10w. Choose ONE of the following options:
 - 50 000 IU once weekly for 6w.
 - 40 000 IU once weekly for 7w.
 - 4 x 1000 IU tablets per day for 10w.
 - 5 x 800 IU capsules per day for 10w.(Do not use combined vitamin D/calcium tablets: will overdose on calcium.)
- **Once loaded, start maintenance therapy** (800–2000 IU vitamin D/d). (Guidance says do this 1m after finishing maintenance, but we are told there isn't much evidence for waiting a month!)
 - **Check corrected calcium 1m after loading with vitamin D** (may unmask primary hyperparathyroidism). If raised, stop vitamin D and investigate.
 - Is calcium intake adequate? Consider using a 'calcium calculator'.

Rapid replacement not needed: start on maintenance therapy

- Give 800–2000 IU vitamin D/d.
- Is calcium intake adequate? Consider the use of a 'calcium calculator'.
- **Check calcium 1m after starting vitamin D** (may unmask primary hyperparathyroidism). If hypercalcaemia, stop vitamin D and investigate.

Rechecking vitamin D levels

Routine vitamin D monitoring not recommended.

Measuring vitamin D may be appropriate if symptomatic vitamin D deficiency or malabsorption, and if suspected poor compliance with medication. If rechecking vitamin D, do so 3–6m after starting treatment.

We make every effort to ensure the information in these pages is accurate and correct at the date of publication, but it is of necessity of a brief and general nature. The information presented herein should not replace your own good clinical judgement, or be regarded as a substitute for taking professional advice in appropriate circumstances. In particular, we suggest you carefully consider the specific facts, circumstances and medical history of any patient, and recommendations of the relevant regulatory authorities. We also suggest that you check drug doses, potential side-effects and interactions with the British National Formulary. Save insofar as any such liability cannot be excluded at law, we do not accept any liability for loss of any type caused by reliance on the information in these pages. June 2025. For full references see the relevant Red Whale articles.